

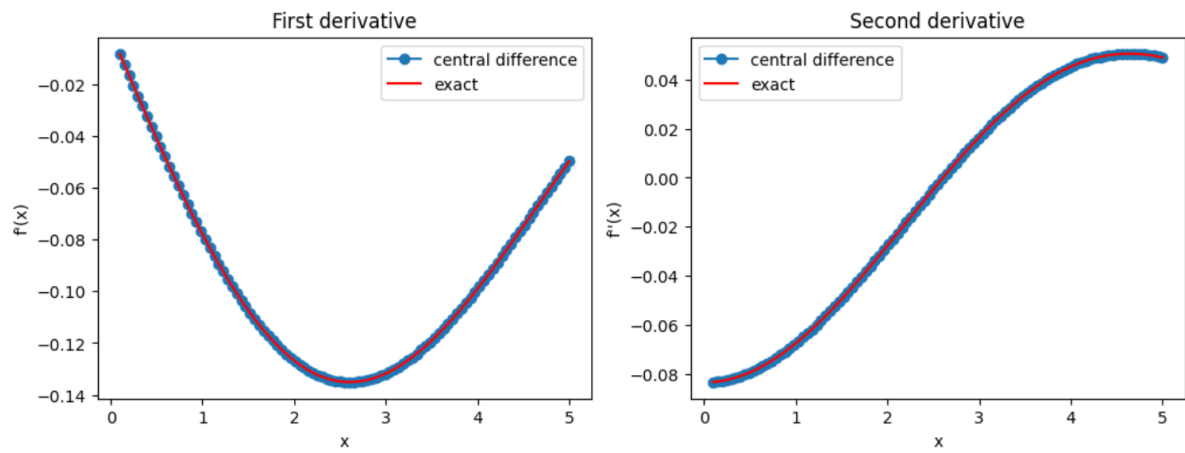
Homework 2

Jason Forbus

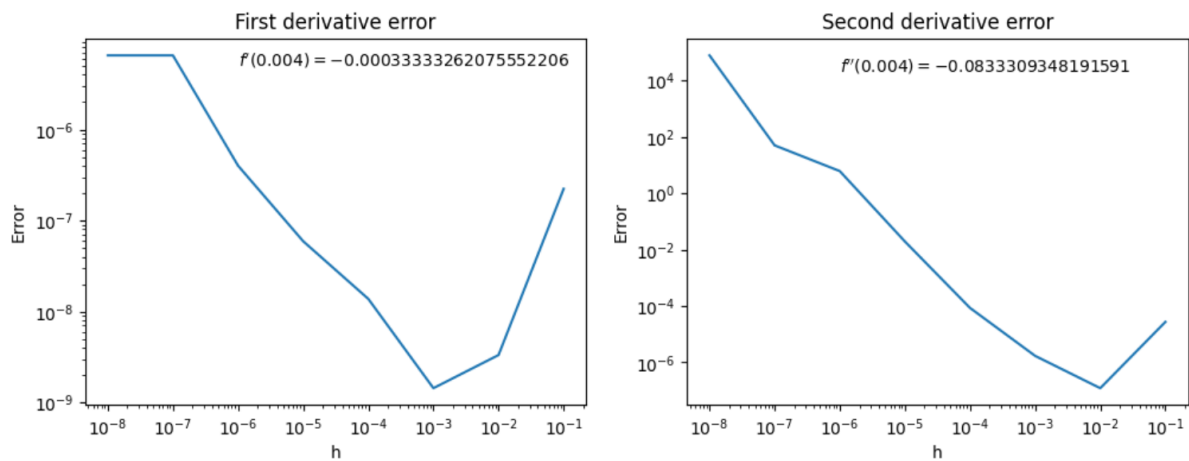
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1

Firstly, I plotted the exact and numerical derivatives together.



This resulted in the errors, with $f'(0.004)$ and $f''(0.004)$ included on the graph:



2

Using Simpson's rule, the integral and error came out to be:

The integral = 4.400000000000001, $h = 0.0002$, $h^4 = 1.6000000000000004e-15$

And the resulting error looked like:

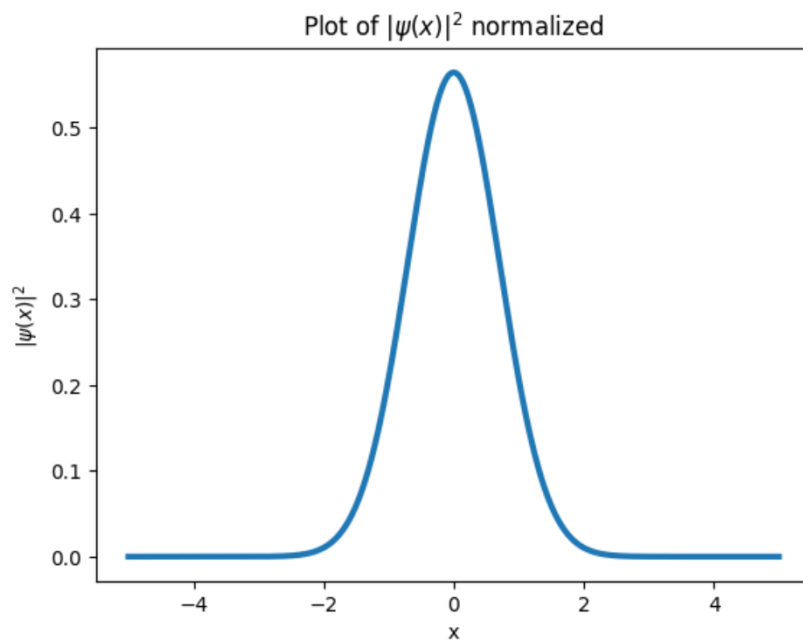


3

3.1

For part a, the normalization constant A and the normalized function came out to be:

$A = 0.7511255444649425$; $h = 0.002$, $h^4 = 1.6000000000000003e-11$



3.2

For part b, the value of B , and the error in the integral, came out to be

$B = 0.6739788945344211$; $h = 0.002$, $h^4 = 1.6000000000000003e-11$

And the probability came out to be

The probability of finding a particle between 0 and 1 = 0.08727721749464779